

ILIAMNA, AK, USA

WMO: 703400

Lat: 59.749N

Lon: 154.909W

Elev: 44

StdP: 100.80

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 25506

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-28.0	-25.0	-32.5	0.2	-26.9	-29.8	0.2	-24.1	15.3	2.0	14.0	-2.3	4.1	350	0.770

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.8	22.3	15.1	20.1	14.2	18.3	13.2	16.1	21.1	14.9	19.0	14.0	17.2	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.9	9.9	17.2	12.9	9.3	16.3	12.2	8.9	15.2	44.8	21.1	41.5	19.0	39.0	17.2	20.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.9	11.4	9.8	-28.7	25.9	3.8	2.4	-31.4	27.6	-33.6	29.0	-35.8	30.4	-38.5	32.1		
			-28.5	17.2	3.3	1.5	-30.9	18.3	-32.8	19.2	-34.6	20.1	-37.0	21.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.6	-7.9	-4.8	-5.1	1.3	6.9	11.2	13.4	13.0	9.3	3.1	-3.4	-6.4	(6)
(7)		DBStd	9.60	9.70	7.84	6.85	4.12	3.03	2.53	2.49	2.34	2.65	4.70	6.89	8.57	(7)
(8)		HDD10.0	2988	554	414	469	261	104	13	1	3	42	216	402	508	(8)
(9)		HDD18.3	5751	813	647	728	511	355	214	154	166	272	473	652	767	(9)
(10)		CDD10.0	283	0	0	0	0	8	49	108	97	20	1	0	0	(10)
(11)		CDD18.3	3	0	0	0	0	0	1	2	1	0	0	0	0	(11)
(12)		CDH23.3	36	0	0	0	0	1	10	22	3	0	0	0	0	(12)
(13)		CDH26.7	4	0	0	0	0	0	1	3	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.3	5.0	4.7	4.1	4.1	3.7	3.7	3.7	4.1	4.7	5.0	5.2	(14)	
(15)	Precipitation	PrecAvg	711	36	28	34	32	39	47	71	121	115	83	55	45	(15)
(16)		PrecMax	1036	72	59	115	67	103	110	168	267	280	203	200	118	(16)
(17)		PrecMin	389	8	4	0	6	0	8	6	52	37	6	2	1	(17)
(18)		PrecStd	179	19	16	28	20	22	31	45	57	67	44	45	29	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	6.5	7.7	12.9	20.8	24.8	26.2	23.5	17.9	12.4	9.4	7.4	(19)
(20)		MCWB	4.1	3.6	4.1	6.7	11.6	15.6	17.3	16.3	12.1	8.7	6.6	5.0	5.0	(20)
(21)		2%	DB	5.7	5.1	5.6	10.2	17.2	21.4	22.7	20.9	15.6	11.2	8.2	6.2	(21)
(22)		MCWB	3.5	2.9	3.0	5.3	9.7	13.6	15.8	15.3	11.3	8.4	5.8	3.8	3.8	(22)
(23)		5%	DB	4.6	4.1	4.1	8.5	14.6	18.7	20.3	19.0	14.3	10.1	7.0	5.1	(23)
(24)		MCWB	2.9	2.2	2.0	4.5	8.7	12.6	14.8	14.1	10.8	7.7	5.0	3.1	3.1	(24)
(25)		10%	DB	3.7	3.2	3.0	6.8	12.2	16.6	18.1	17.2	13.2	9.1	5.6	3.9	(25)
(26)		MCWB	2.2	1.5	1.2	3.6	7.5	11.5	13.7	13.3	10.4	6.9	3.8	2.3	2.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.1	4.1	4.7	7.5	12.4	16.3	18.2	17.6	13.3	10.1	7.3	5.5	(27)
(28)		MDCB	6.9	5.7	6.9	12.2	19.5	23.5	24.6	21.5	16.0	11.5	8.6	6.8	6.8	(28)
(29)		2%	WB	3.8	3.2	3.4	5.7	10.4	14.4	16.4	15.9	12.4	8.9	6.4	4.4	(29)
(30)		MDCB	5.3	4.7	5.2	9.4	16.0	19.9	22.1	19.8	14.3	10.6	8.0	5.9	5.9	(30)
(31)		5%	WB	3.0	2.4	2.3	4.7	9.0	13.1	15.1	14.7	11.6	8.0	5.1	3.2	(31)
(32)		MDCB	4.4	3.8	3.8	7.9	13.9	17.6	19.5	18.0	13.4	9.8	6.9	4.6	4.6	(32)
(33)		10%	WB	2.2	1.7	1.5	3.9	7.9	11.9	14.1	13.8	10.8	7.2	3.8	2.3	(33)
(34)		MDCB	3.6	3.0	2.8	6.5	11.7	15.8	17.4	16.4	12.7	9.0	5.4	3.8	3.8	(34)
(35)	Mean Daily Temperature Range	MDBR	6.2	6.4	7.7	7.3	8.0	8.2	6.8	6.6	6.2	5.7	5.9	6.1	(35)	
(36)		5% DB	MDCDBR	4.8	4.7	6.3	9.6	13.1	13.0	11.5	10.1	8.0	5.2	4.5	4.6	(36)
(37)		MCWBR	3.6	3.4	4.4	5.8	7.1	6.5	5.7	5.4	4.7	3.6	3.3	3.5	3.5	(37)
(38)		5% WB	MDCDBR	4.5	4.5	5.3	8.6	11.6	11.6	10.4	8.7	6.4	4.4	4.5	4.6	(38)
(39)	MCWBR	3.4	3.4	3.9	5.4	6.4	6.0	5.4	4.9	4.3	3.3	3.6	3.8	3.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.245	0.283	0.306	0.385	0.383	0.376	0.374	0.358	0.326	0.302	0.260	0.246	(40)	
(41)		taud	2.137	2.244	2.208	2.066	2.259	2.365	2.420	2.458	2.523	2.415	2.098	1.956	(41)	
(42)		Ebn at Noon	545	728	823	802	842	856	849	837	811	706	494	348	(42)	
(43)		Edh at Noon	43	69	97	135	120	111	102	92	73	58	44	34	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.37	0.97	2.60	3.98	4.77	5.26	4.42	3.48	2.38	1.21	0.49	0.22	(44)	
(45)		RadStd	0.05	0.19	0.40	0.49	0.52	0.45	0.40	0.44	0.35	0.11	0.07	0.02	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (11 neighbors)	N/A	+3.16	+2.78	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		N/A	+2.40	+1.92	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page